

## **1. Introduction**

### **Aim**

The primary aim of FitBuddy – AI-Powered Personal Workout Coach & Adaptive Nutrition Planner is to develop an intelligent fitness application that provides personalized workout plans, adaptive nutrition guidance, wellness monitoring, and AI-based coaching. The system uses Artificial Intelligence to understand each user's physical profile, fitness goals, and daily motivation level to deliver customized fitness recommendations.

The project also aims to help users develop healthy habits by providing guided workout sessions, real-time workout timers, wellness tracking, and continuous progress monitoring. By integrating AI with fitness management, FitBuddy offers an affordable and accessible alternative to traditional personal fitness coaching.

### **About the Project**

Maintaining a healthy lifestyle has become increasingly difficult due to busy work schedules, academic pressure, unhealthy eating habits, and limited access to professional fitness guidance. Although many fitness applications are available today, most provide generic workout routines and fixed diet plans that do not consider the unique needs of individual users. As a result, users often lose motivation and discontinue their fitness journey before achieving meaningful results.

FitBuddy is an AI-powered fitness and wellness platform designed to overcome these limitations by providing personalized recommendations based on the user's physical profile, fitness goals, daily motivation, and wellness status. The application uses Google Gemini AI to generate customized workout schedules and adaptive nutrition plans. Instead of following the same routine every day, the system intelligently adjusts recommendations according to the user's current condition and progress.

One of the unique features of the project is the Josh Meter, which measures the user's daily motivation level. Based on whether the motivation is low, medium, or high, the system automatically adjusts workout intensity. For example, users with low motivation may receive light stretching or short exercise routines, while highly motivated users receive more intensive workout plans.

FitBuddy also includes a Start Workout feature that allows users to begin guided workout sessions directly within the application. A built-in Workout Timer manages exercise duration and rest intervals, helping users perform workouts efficiently and maintain proper timing.

To support overall health management, the application provides a Wellness Tracker that records daily water intake, sleep duration, mood, weight, BMI, and other wellness activities. These features encourage users to build healthy habits beyond exercise alone.

Another important component is the AI Coach, which acts as a virtual personal trainer. Users can ask fitness-related questions, receive exercise guidance, obtain nutrition advice, and get motivational support at any time. The AI Coach provides intelligent responses using Gemini AI, making professional fitness guidance more accessible.

The application continuously stores user data, workout history, nutrition plans, and wellness information in the database. Based on the user's progress, the AI system updates future workout and meal recommendations, ensuring that the guidance evolves with the user's fitness journey.

Overall, FitBuddy combines workout planning, nutrition management, wellness tracking, guided exercise sessions, AI coaching, and progress monitoring into a single platform. The project aims to make fitness more personalized, engaging, and effective for users of all fitness levels.

### Key Features

- AI-powered personalized workout planning.
- Adaptive nutrition recommendations using Gemini AI.
- Josh Meter for daily motivation analysis.
- Guided workout sessions with Start Workout.
- Built-in workout and rest timer.
- Wellness Tracker for sleep, water intake, mood, and BMI.
- AI Coach for fitness and nutrition guidance.
- Progress tracking with personalized recommendations.
- User authentication and profile management.

- Interactive dashboard with reports and analytics.

### **Objectives**

- Develop an AI-powered fitness coaching application.
- Generate personalized workout plans based on user goals.
- Provide adaptive nutrition planning using AI.
- Monitor overall wellness through health tracking.
- Improve user motivation using the Josh Meter.
- Guide users through workout sessions with timers.
- Track user progress and continuously improve recommendations.
- Encourage healthy lifestyle habits through intelligent coaching.

### **Advantages**

- Personalized fitness experience.
- Dynamic workout intensity adjustment.
- Intelligent nutrition planning.
- Real-time workout guidance.
- Comprehensive wellness monitoring.
- AI-powered virtual fitness coach.
- Improved motivation and consistency.
- Easy-to-use and responsive interface.
- Affordable alternative to personal trainers.
- Continuous progress tracking and reporting.

## **2. LITERATURE SURVEY**

### **EXISTING SYSTEM**

In recent years, many fitness applications have been developed to help users maintain a healthy lifestyle. Popular applications such as MyFitnessPal, Nike Training Club, Google Fit, Fitbit, and HealthifyMe provide features like workout plans, calorie tracking, activity monitoring, and diet recommendations. These applications have helped many users improve their fitness by offering digital health management solutions.

However, most existing systems have several limitations. They usually provide predefined workout routines and general diet plans that are the same for all users. These systems rarely consider the user's daily motivation, changing energy levels, or individual fitness progress. As a result, users often lose interest and discontinue their fitness journey.

Most applications also focus on only one aspect of fitness, such as exercise tracking or calorie counting, rather than providing a complete solution that combines workout planning, nutrition, wellness monitoring, and intelligent coaching.

### **Limitations of Existing System**

- Generic workout and diet plans for all users.
- Limited personalization based on individual fitness goals.
- No daily motivation analysis.
- Workout intensity remains mostly fixed.
- Limited wellness monitoring features.
- Lack of AI-powered coaching and guidance.
- Few applications provide integrated workout timers.
- Minimal adaptive recommendations based on user progress.

- Dependence on manual planning by users.
- Limited interaction and motivation support.

## **PROPOSED SYSTEM**

The proposed system, FitBuddy – AI-Powered Personal Workout Coach & Adaptive Nutrition Planner, is designed to overcome the limitations of existing fitness applications by providing intelligent, adaptive, and personalized fitness guidance.

FitBuddy uses Google Gemini AI to generate customized workout schedules and nutrition plans according to each user's profile, including age, height, weight, BMI, activity level, and fitness goals. Unlike traditional applications, FitBuddy continuously adjusts recommendations based on user progress and daily motivation.

A unique feature of the system is the Josh Meter, which measures the user's daily motivation level. Based on the motivation score, the application automatically adjusts workout intensity, ensuring that users receive appropriate exercise recommendations according to their current energy level.

The application also includes several innovative modules such as:

- AI Workout Planner
- AI Nutrition Planner
- Start Workout
- Workout Timer
- Wellness Tracker
- AI Coach
- Progress Tracker
- Personalized Dashboard

The Start Workout module guides users through their exercise sessions, while the integrated Workout Timer manages exercise duration and rest intervals. The Wellness Tracker records sleep, water intake, mood, weight, and BMI, helping users maintain overall health.

The AI Coach acts as a virtual fitness trainer by answering fitness-related questions, providing exercise guidance, offering nutrition advice, and motivating users throughout their fitness journey.

By combining AI-powered recommendations with wellness monitoring and intelligent coaching, FitBuddy provides a complete digital fitness companion suitable for users of different age groups and fitness levels.

### Technologies used

The proposed system is developed using modern web technologies and artificial intelligence tools to ensure high performance, scalability, and user-friendly interaction.

| Technology       | Purpose                                 |
|------------------|-----------------------------------------|
| Next.js          | Frontend and Backend Development        |
| React.js         | User Interface Development              |
| TypeScript       | Type-safe Application Development       |
| Tailwind CSS     | Responsive User Interface Design        |
| Node.js          | Server-side Processing                  |
| Prisma ORM       | Database Management                     |
| PostgreSQL       | Data Storage                            |
| NextAuth.js      | User Authentication                     |
| Google Gemini AI | AI Workout, Nutrition & Coaching        |
| REST APIs        | Communication between Client and Server |
| Vercel           | Deployment Platform                     |
| Git & GitHub     | Version Control                         |

### **3. HARDWARE AND SOFTWARE REQUIREMENTS**

This chapter describes the hardware and software required to develop and run the FitBuddy – AI-Powered Personal Workout Coach & Adaptive Nutrition Planner application. The selected technologies ensure that the system is scalable, secure, user-friendly, and capable of providing AI-powered fitness recommendations.

#### **HARDWARE SYSTEM CONFIGURATION**

The following hardware specifications are recommended for the successful development and execution of the FitBuddy application.

##### **Development System Requirements**

| <b>Hardware Component</b> | <b>Minimum Requirement</b> | <b>Recommended Requirement</b>  |
|---------------------------|----------------------------|---------------------------------|
| Processor                 | Intel Core i3 (8th Gen)    | Intel Core i5/i7 or AMD Ryzen 5 |
| RAM                       | 4 GB                       | 8 GB or Above                   |
| Storage                   | 50 GB Free Space           | 256 GB SSD or Above             |
| Display                   | 1366 × 768                 | Full HD (1920 × 1080)           |
| Internet                  | Broadband Connection       | High-Speed Internet             |
| Keyboard & Mouse          | Standard                   | Standard                        |
| Smartphone                | Android/iOS (Optional)     | Android/iOS (Latest Version)    |

##### **Server Requirements**

| <b>Component</b> | <b>Specification</b>             |
|------------------|----------------------------------|
| Cloud Platform   | Vercel                           |
| Database Server  | PostgreSQL                       |
| Authentication   | NextAuth.js                      |
| AI Service       | Google Gemini API                |
| Storage          | Cloudinary (Optional for Images) |

## SOFTWARE SYSTEM CONFIGURATION

The FitBuddy application is developed using modern web technologies to ensure performance, maintainability, and scalability.

| Software                | Version / Technology   | Purpose                             |
|-------------------------|------------------------|-------------------------------------|
| Operating System        | Windows 10/11          | Development Environment             |
| Frontend                | Next.js 15             | User Interface Development          |
| Programming Language    | TypeScript             | Type-safe Development               |
| UI Framework            | React.js               | Component-based UI                  |
| Styling                 | Tailwind CSS           | Responsive Design                   |
| Backend                 | Next.js Route Handlers | REST API Development                |
| Database                | PostgreSQL             | Data Storage                        |
| ORM                     | Prisma ORM             | Database Management                 |
| Authentication          | NextAuth.js            | Secure Login                        |
| Artificial Intelligence | Google Gemini AI       | Workout & Nutrition Recommendations |
| API Testing             | Postman                | API Testing                         |
| Code Editor             | Visual Studio Code     | Development                         |
| Version Control         | Git & GitHub           | Source Code Management              |
| Browser                 | Google Chrome          | Testing                             |



## **4. SOFTWARE REQUIREMENT SPECIFICATION (SRS)**

This chapter defines the software requirements for FitBuddy – AI-Powered Personal Workout Coach & Adaptive Nutrition Planner. It describes the purpose, scope, functional requirements, non-functional requirements, user characteristics, constraints, and assumptions of the system. The SRS acts as a blueprint for designing, developing, testing, and maintaining the application.

### **INTRODUCTION**

Software Requirement Specification (SRS) is a document that describes all the functional and non-functional requirements of a software system. It provides a detailed understanding of how the system should operate and what services it should provide.

The SRS for FitBuddy defines the requirements for an AI-powered fitness platform that generates personalized workout plans, adaptive nutrition recommendations, wellness monitoring, and AI coaching. The system is designed to help users achieve their fitness goals through intelligent recommendations based on their profile, fitness objectives, and daily motivation level.

### **PURPOSE**

The main purpose of FitBuddy is to provide users with a complete digital fitness platform that combines artificial intelligence, health monitoring, and personalized coaching.

The objectives of the system are:

- To provide personalized workout plans using AI.
- To generate adaptive nutrition recommendations.
- To monitor user wellness and fitness progress.
- To help users stay motivated through the Josh Meter.
- To provide guided workout sessions with a built-in timer.
- To offer AI-based coaching and fitness guidance.

- To improve consistency and overall health through personalized recommendations.

## SCOPE

The scope of FitBuddy includes the development of an intelligent web application that helps users manage their fitness journey efficiently.

The application allows users to:

- Register and securely log in.
- Create and manage personal fitness profiles.
- Select fitness goals such as weight loss, muscle gain, endurance, or weight gain.
- Generate personalized workout plans.
- Receive AI-generated nutrition recommendations.
- Start guided workout sessions.
- Use workout and rest timers.
- Track wellness activities such as water intake, sleep, mood, weight, and BMI.
- Interact with an AI Coach for fitness-related guidance.
- Monitor progress using dashboards and reports.

Future versions may support wearable device integration, voice-based coaching, and real-time posture detection.

## **FUNCTIONAL REQUIREMENTS**

Functional requirements describe what the system should do.

### **1. User Authentication Module**

The system shall:

- Allow user registration.
- Allow secure login.
- Support Google Sign-In.
- Provide password recovery.
- Manage user sessions securely.

### **2. User Profile Module**

The system shall allow users to:

- Update personal information.
- Enter age, height, weight, and gender.
- Select activity level.
- Choose fitness goals.
- Update profile information.

### **3. Workout Planner Module**

The system shall:

- Generate AI-based workout plans.
- Recommend exercises according to fitness goals.

- Display sets, repetitions, and duration.
- Modify workout intensity using the Josh Meter.
- Save workout history.

#### **4. Nutrition Planner Module**

The system shall:

- Generate personalized meal plans.
- Recommend daily calorie intake.
- Suggest protein, carbohydrate, and fat intake.
- Display breakfast, lunch, dinner, and snacks.
- Update meal plans based on progress.

#### **5. Josh Meter Module**

The system shall:

- Measure daily motivation.
- Categorize motivation as Low, Medium, or High.
- Adjust workout intensity automatically.
- Provide motivational suggestions.

#### **6. Start Workout Module**

The system shall:

- Start workout sessions.
- Display current exercise.

- Mark completed exercises.
- Calculate workout duration.
- Save workout progress.

## **7. Workout Timer Module**

The system shall:

- Start countdown timers.
- Pause and resume timers.
- Set exercise duration.
- Manage rest intervals.
- Notify users when time is completed.

## **8. Wellness Tracker Module**

The system shall:

- Record water intake.
- Track sleep duration.
- Monitor daily mood.
- Store body weight.
- Calculate BMI.
- Display wellness history.

## **9. AI Coach Module**

The system shall:

- Answer fitness questions.
- Recommend exercises.
- Suggest healthy meals.
- Explain exercise techniques.
- Provide motivational guidance.
- Give recovery advice.

## **10. Progress Tracker Module**

The system shall:

- Track completed workouts.
- Monitor body weight.
- Display BMI progress.
- Show calorie statistics.
- Generate weekly and monthly reports.

## **4.5 Non-Functional Requirements**

### **Performance**

- Fast page loading.
- AI responses within a few seconds.
- Smooth user experience.

- Efficient database operations.

### **Reliability**

- Stable application performance.
- Secure data storage.
- Accurate workout recommendations.
- Reliable authentication.

### **Security**

- User authentication.
- Password encryption.
- Protected API routes.
- Secure database access.
- Role-based authorization.

### **Usability**

- Simple user interface.
- Easy navigation.
- Mobile responsive design.
- Beginner-friendly experience.

### **Scalability**

The application should support:

- Large numbers of users.

- Future feature expansion.
- Additional AI capabilities.
- Cloud deployment.

### **Maintainability**

The application should have:

- Modular code structure.
- Reusable components.
- Proper documentation.
- Easy debugging and updates.

### **Portability**

The system should run on:

- Windows
- Linux
- macOS

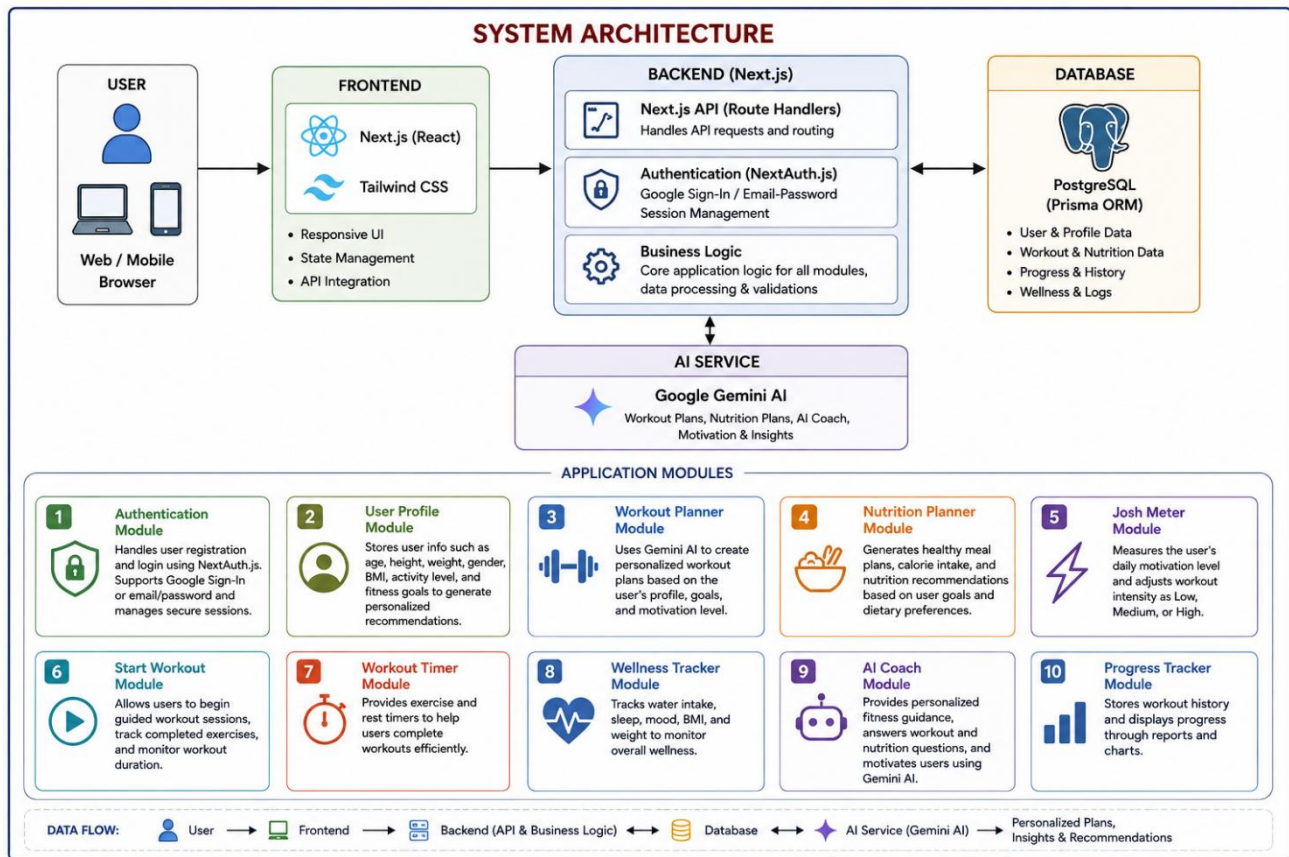
Supported browsers:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari



## 5. SYSTEM DESIGN

### 5.1 System Architecture



The FitBuddy system follows a client-server architecture. Users interact with the application through a web interface developed using Next.js. The backend processes user requests and communicates with the PostgreSQL database using Prisma ORM. Google Gemini AI generates personalized workout plans, nutrition recommendations, and AI coaching. The system also manages workout sessions, wellness tracking, and progress monitoring.

### 5.2 Modules

#### 1. Authentication Module

Handles user registration and login using NextAuth.js. It authenticates users through Google Sign-In or email/password and manages secure sessions.

## **2. User Profile Module**

Stores user information such as age, height, weight, gender, BMI, activity level, and fitness goals. This information is used to generate personalized recommendations.

## **3. Workout Planner Module**

Uses Gemini AI to create personalized workout plans according to the user's profile, goals, and motivation level.

## **4. Nutrition Planner Module**

Generates healthy meal plans, calorie intake, and nutrition recommendations based on user requirements.

## **5. Josh Meter Module**

Measures the user's daily motivation level and adjusts workout intensity as Low, Medium, or High.

## **6. Start Workout Module**

Allows users to begin guided workout sessions, track completed exercises, and monitor workout duration.

## **7. Workout Timer Module**

Provides exercise and rest timers to help users complete workouts efficiently.

## **8. Wellness Tracker Module**

Tracks water intake, sleep, mood, BMI, and weight to monitor overall wellness.

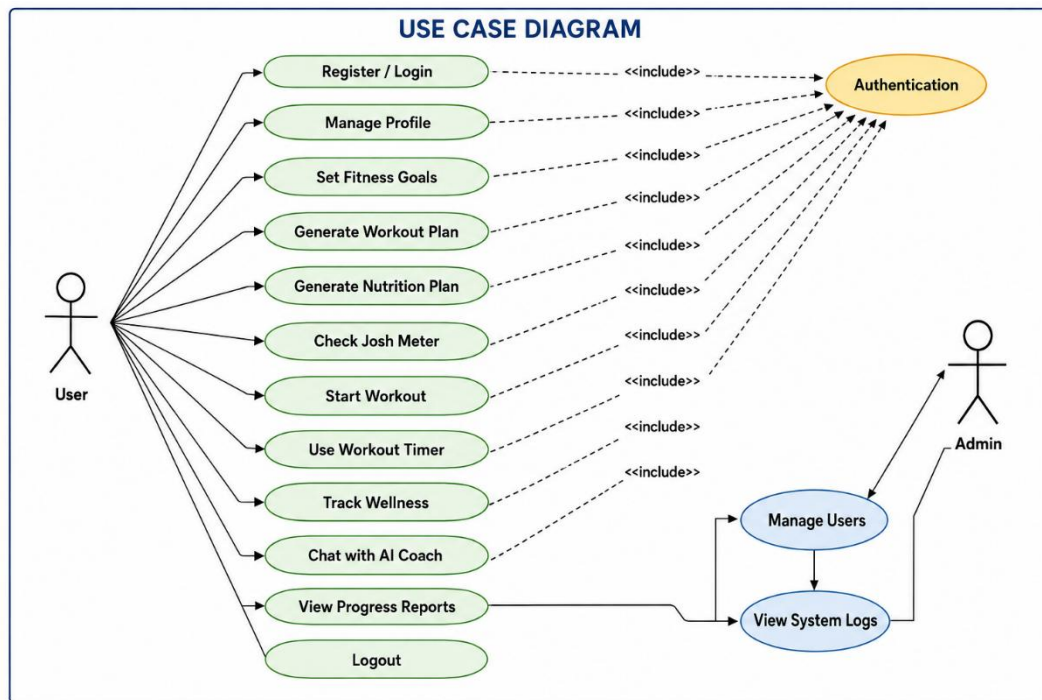
## **9. AI Coach Module**

Provides personalized fitness guidance, answers workout and nutrition questions, and motivates users using Gemini AI.

## **10. Progress Tracker Module**

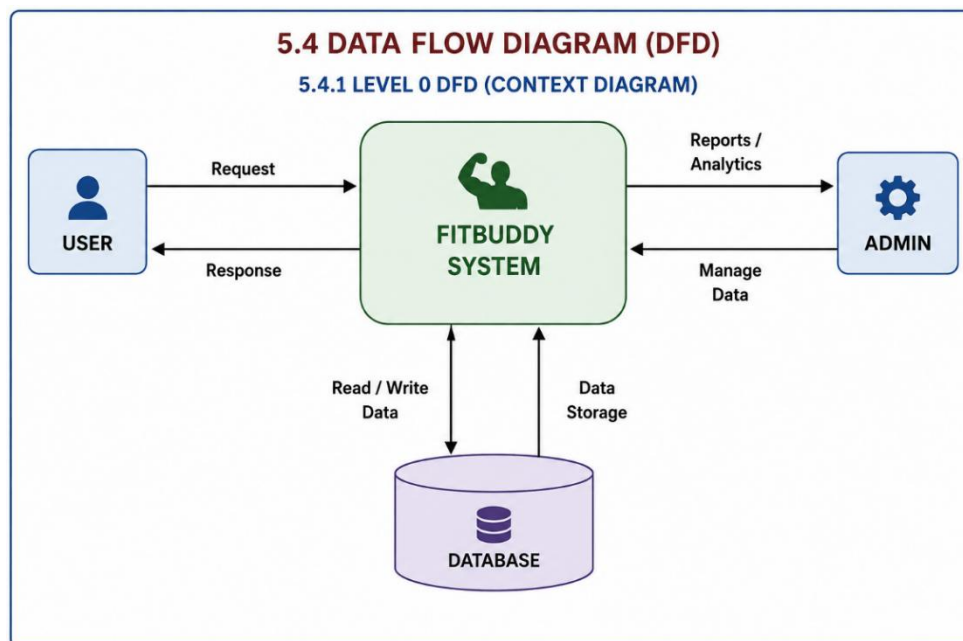
Stores workout history and displays progress through reports and charts.

### 5.3 Use Case Diagram



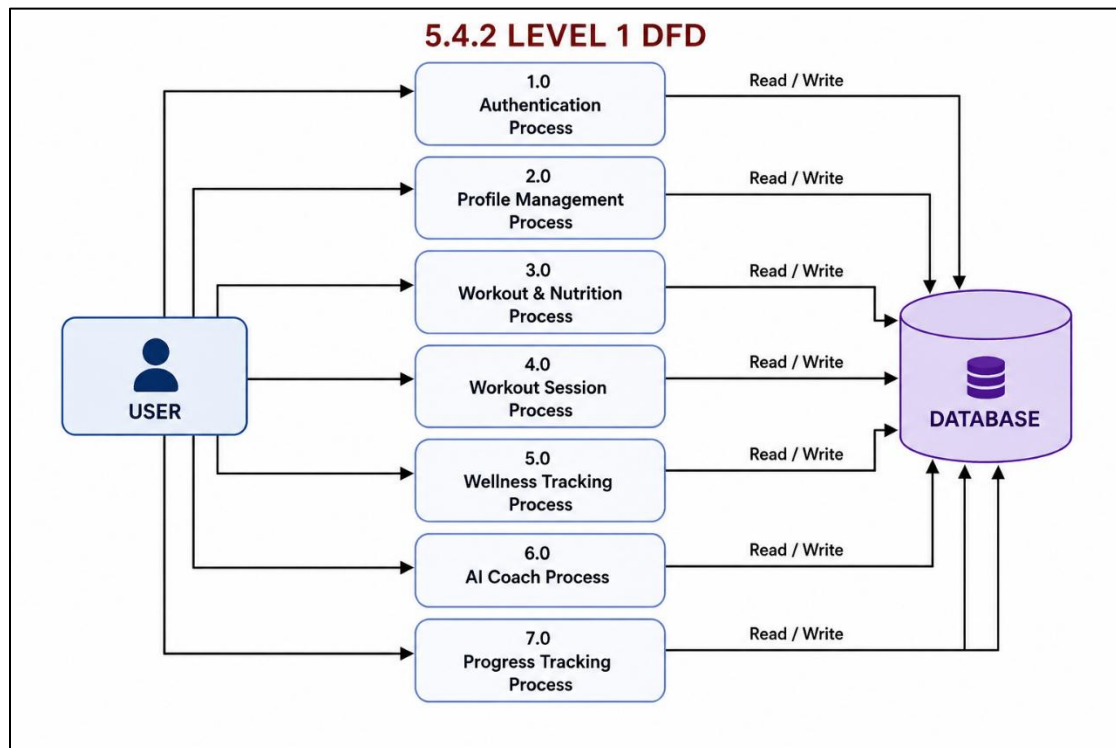
The use case diagram illustrates the interaction between the User and the FitBuddy system. The user can register, log in, manage the profile, generate workout and nutrition plans, start workouts, use the workout timer, track wellness, communicate with the AI Coach, and monitor fitness progress.

### 5.4 Data Flow Diagram



### 5.4.1 Level 0 DFD (Context Diagram)

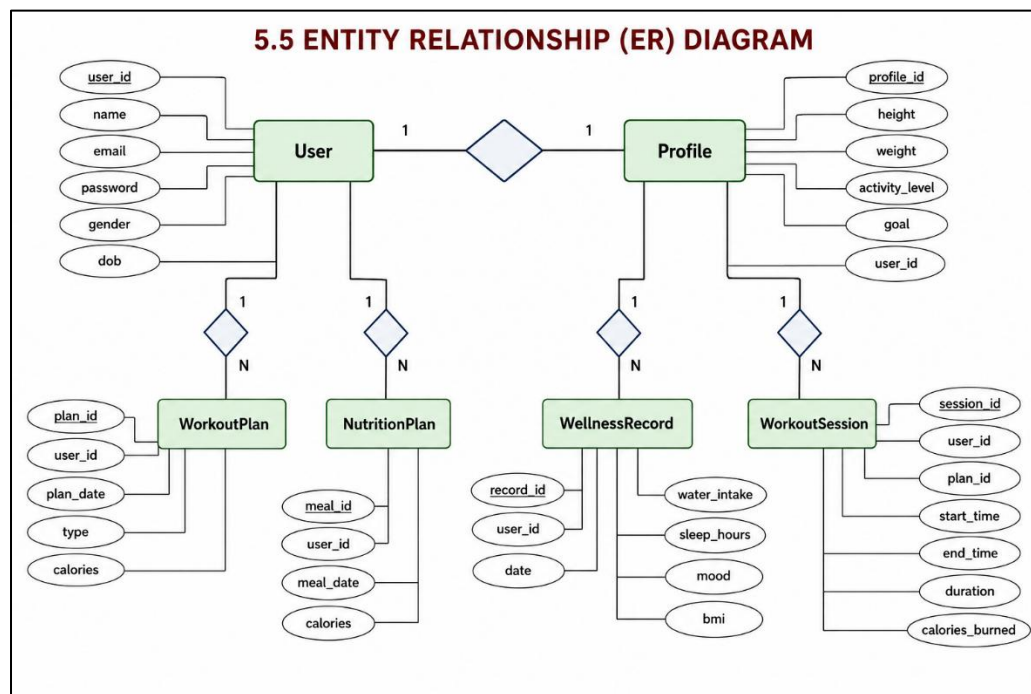
The Level 0 Data Flow Diagram provides a high-level overview of the FitBuddy system and its interaction with external entities. The user sends requests such as login, profile updates, workout planning, and wellness tracking to the system, which processes them and returns appropriate responses. The system communicates with the database to store and retrieve user information and fitness records. The admin manages system data and receives reports and analytics for monitoring and maintenance.



### 5.4.2 Level 1 DFD

The Level 1 Data Flow Diagram illustrates the internal functional processes of the FitBuddy system in greater detail. It divides the system into key modules such as Authentication, Profile Management, Workout & Nutrition, Workout Session, Wellness Tracking, AI Coach, and Progress Tracking. Each module interacts with the database to read and write the required data while processing user requests. This diagram clearly represents the flow of information between the user, system processes, and the database for efficient application functionality.

## 5.5 ER Diagram



The ER diagram represents the relationship between entities such as User, Profile, Workout Plan, Nutrition Plan, Wellness Record, Progress Report, Josh Meter, and AI Coach. These entities are connected through primary and foreign keys to ensure proper data management.

## 5.6 Database Tables

### 5.6 DATABASE TABLES

| No. | Table Name              | Description / Purpose           | Key Fields (Primary Key in <b>Bold</b> , Foreign Key in <i>Italics</i> )                                      |
|-----|-------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1   | <b>users</b>            | Stores user account information | <b>user_id</b> , name, email, password, gender, dob, created_at                                               |
| 2   | <b>profiles</b>         | Stores fitness profile          | <b>profile_id</b> , <i>user_id</i> , height, weight, activity_level, goal                                     |
| 3   | <b>workout_plans</b>    | Stores workout plans            | <b>plan_id</b> , <i>user_id</i> , plan_date, type, intensity, calories, plan_data                             |
| 4   | <b>nutrition_plans</b>  | Stores meal plans               | <b>meal_id</b> , <i>user_id</i> , meal_date, meal_type, calories, protein, carbs, fats                        |
| 5   | <b>workout_sessions</b> | Stores workout history          | <b>session_id</b> , <i>user_id</i> , <i>plan_id</i> , start_time, end_time, duration, calories_burned, status |
| 6   | <b>wellness_records</b> | Stores sleep, water, mood       | <b>record_id</b> , <i>user_id</i> , record_date, water_intake, sleep_hours, mood, weight, bmi, notes          |
| 7   | <b>progress_reports</b> | Stores fitness progress         | <b>report_id</b> , <i>user_id</i> , report_type, period, data_json, created_at                                |
| 8   | <b>josh_meter</b>       | Stores motivation level         | <b>josh_id</b> , <i>user_id</i> , date, motivation_level, score, suggestions                                  |
| 9   | <b>ai_chat</b>          | Stores AI Coach conversations   | <b>chat_id</b> , <i>user_id</i> , message, response, created_at                                               |

## **6. SYSTEM IMPLEMENTATION**

### **6.1 Introduction**

System implementation is the process of converting the designed system into a working software application. In this phase, all modules of the FitBuddy application are developed, integrated, and tested to ensure smooth functionality. The application is built using modern web technologies and Artificial Intelligence to provide users with personalized fitness guidance.

### **6.2 Development Environment**

| <b>Component</b> | <b>Technology Used</b>           |
|------------------|----------------------------------|
| Frontend         | Next.js 15, React.js, TypeScript |
| Styling          | Tailwind CSS, Shaden UI          |
| Backend          | Next.js Route Handlers           |
| Database         | PostgreSQL                       |
| ORM              | Prisma ORM                       |
| Authentication   | NextAuth.js                      |
| AI               | Google Gemini AI                 |
| Charts           | Recharts                         |
| Deployment       | Vercel                           |
| Version Control  | Git & GitHub                     |

### **6.3 Implementation of Modules**

#### **1. User Authentication**

The application provides secure user authentication using NextAuth.js. Users can register using email and password or log in using their Google account. Authentication ensures that only authorized users can access their fitness data and personalized recommendations.



## **2. User Profile Management**

Users enter personal information such as age, height, weight, gender, activity level, and fitness goals. The system stores this information in the database and uses it to generate personalized workout and nutrition plans.

## **3. AI Workout Planner**

The Workout Planner uses Google Gemini AI to generate personalized workout schedules based on the user's profile, fitness goal, and Josh Meter score. The generated plan includes exercises, sets, repetitions, duration, and difficulty level.

## **4. AI Nutrition Planner**

The Nutrition Planner creates personalized meal plans according to the user's calorie requirements and fitness objectives. It recommends balanced meals, water intake, and daily nutritional requirements.

## **5. Josh Meter**

The Josh Meter measures the user's daily motivation level (Low, Medium, or High). Based on the selected motivation level, the system adjusts workout intensity and recommends suitable exercises.

## **6. Start Workout**

The Start Workout feature allows users to begin guided workout sessions. During the workout, users can follow exercise instructions and mark completed exercises. This helps maintain consistency and discipline.

## **7. Workout Timer**

The Workout Timer manages exercise and rest intervals. Users can start, pause, resume, or reset the timer. Notifications are displayed when the exercise or rest period is completed.

## **8. Wellness Tracker**

The Wellness Tracker records important health information such as water intake, sleep duration, mood, weight, and BMI. These records help users monitor their overall health and maintain healthy habit

## 9. AI Coach

The AI Coach acts as a virtual fitness assistant powered by Google Gemini AI. It answers user questions, provides workout suggestions, nutrition advice, and motivational guidance based on the user's fitness goals.

## 10. Progress Tracker

The Progress Tracker stores workout history, nutrition records, wellness data, and fitness achievements. It displays progress through charts and reports, allowing users to monitor improvements over time.

### 6.4 Database Implementation

The application uses PostgreSQL as the database and Prisma ORM for database management.

The main tables include:

- Users
- Profiles
- Workout Plans
- Nutrition Plans
- Workout Sessions
- Wellness Records
- Josh Meter
- Progress Reports
- AI Chat History

These tables are connected using primary and foreign keys to ensure data consistency and integrity.



## 6.5 AI Integration

Google Gemini AI is integrated into the application through REST APIs. User profile details, fitness goals, and motivation level are sent to Gemini AI, which generates personalized workout plans, nutrition recommendations, and AI Coach responses. The generated results are displayed on the user dashboard and stored for future reference.

## 6.6 User Interface

The user interface is developed using Next.js, React.js, Tailwind CSS, and Shaden UI.

Key UI features include:

- Responsive design
- Light and dark mode support
- Interactive dashboard
- Easy navigation
- Modern card-based layout
- Mobile-friendly interface

## 6.7 Security Implementation

The application implements several security measures to protect user data:

- Secure authentication using NextAuth.js
- Password encryption
- Protected API routes
- Session management
- Input validation

- Secure database communication

## **7. SYSTEM TESTING**

### **7.1 Introduction**

System testing is an important phase in software development that ensures the application functions correctly and meets the specified requirements. The FitBuddy application was tested to verify that all modules work properly, user interactions are handled correctly, and the system provides accurate and reliable results.

The testing process focused on validating user authentication, workout generation, nutrition planning, AI coaching, wellness tracking, workout timer functionality, and overall system performance.

### **7.2 Objectives of Testing**

The objectives of testing are:

- To verify that all modules work correctly.
- To identify and fix software defects.
- To ensure data accuracy and consistency.
- To validate AI-generated recommendations.
- To improve application performance and reliability.
- To provide a smooth user experience.

### **7.3 Types of Testing Performed**

#### **1. Unit Testing**

Each module was tested individually to ensure it performs its intended functionality.

Examples:

- User Login

- Registration
- Workout Planner
- Nutrition Planner
- Josh Meter
- Workout Timer
- AI Coach

## **2. Integration Testing**

After individual modules were tested, they were integrated to verify smooth communication between different modules.

Examples:

- Login → Dashboard
- Profile → AI Workout
- Workout → Progress Tracker
- Wellness Tracker → Dashboard
- AI Coach → User Profile

## **3. System Testing**

The complete application was tested as a whole to ensure that all features work together correctly.

The following functionalities were verified:

- Authentication
- AI Recommendations

- Workout Sessions
- Wellness Tracking
- Dashboard
- Reports

#### 4. User Acceptance Testing (UAT)

The application was tested from the user's perspective to ensure it satisfies user requirements.

The feedback confirmed:

- Easy navigation
- User-friendly interface
- Accurate workout recommendations
- Responsive design
- Smooth overall performance

#### 7.4 Test Cases

| Test Case ID | Test Scenario           | Expected Result                       | Status |
|--------------|-------------------------|---------------------------------------|--------|
| TC01         | User Registration       | Account created successfully          | Pass   |
| TC02         | User Login              | User logged into dashboard            | Pass   |
| TC03         | Update Profile          | Profile updated successfully          | Pass   |
| TC04         | Generate Workout Plan   | Personalized workout displayed        | Pass   |
| TC05         | Generate Nutrition Plan | Meal plan generated correctly         | Pass   |
| TC06         | Josh Meter Selection    | Workout intensity adjusted            | Pass   |
| TC07         | Start Workout           | Workout session started               | Pass   |
| TC08         | Workout Timer           | Timer started and completed correctly | Pass   |
| TC09         | Wellness Tracker        | Wellness data saved successfully      | Pass   |

|      |                  |                                       |      |
|------|------------------|---------------------------------------|------|
| TC10 | AI Coach         | AI provided relevant fitness guidance | Pass |
| TC11 | Progress Tracker | Progress displayed correctly          | Pass |
| TC12 | Logout           | User logged out securely              | Pass |

### 7.5 Performance Testing

The application was tested for performance under normal usage conditions.

| Parameter            | Result     |
|----------------------|------------|
| Login Response       | Successful |
| Dashboard Loading    | Fast       |
| Workout Generation   | Fast       |
| Nutrition Generation | Fast       |
| Database Access      | Stable     |
| AI Response Time     | Acceptable |
| Overall Performance  | Good       |

### 7.6 Security Testing

The following security checks were performed:

- User authentication validation
- Protected routes
- Password encryption
- Session management
- Unauthorized access prevention
- Input validation
- Secure API communication

Result: No major security issues were observed during testing.

## 7.7 Compatibility Testing

The application was tested on different browsers.

| Browser         | Result |
|-----------------|--------|
| Google Chrome   | Pass   |
| Microsoft Edge  | Pass   |
| Mozilla Firefox | Pass   |
| Brave Browser   | Pass   |

The responsive interface was also verified on desktop and mobile screen sizes.

## 7.8 Testing Summary

| Module            | Status |
|-------------------|--------|
| Authentication    | Pass   |
| User Profile      | Pass   |
| Workout Planner   | Pass   |
| Nutrition Planner | Pass   |
| Josh Meter        | Pass   |
| Start Workout     | Pass   |
| Workout Timer     | Pass   |
| Wellness Tracker  | Pass   |
| AI Coach          | Pass   |
| Progress Tracker  | Pass   |
| Dashboard         | Pass   |

## 7.9 Advantages of Testing

- Improved software quality.
- Reduced application errors.
- Better user experience.

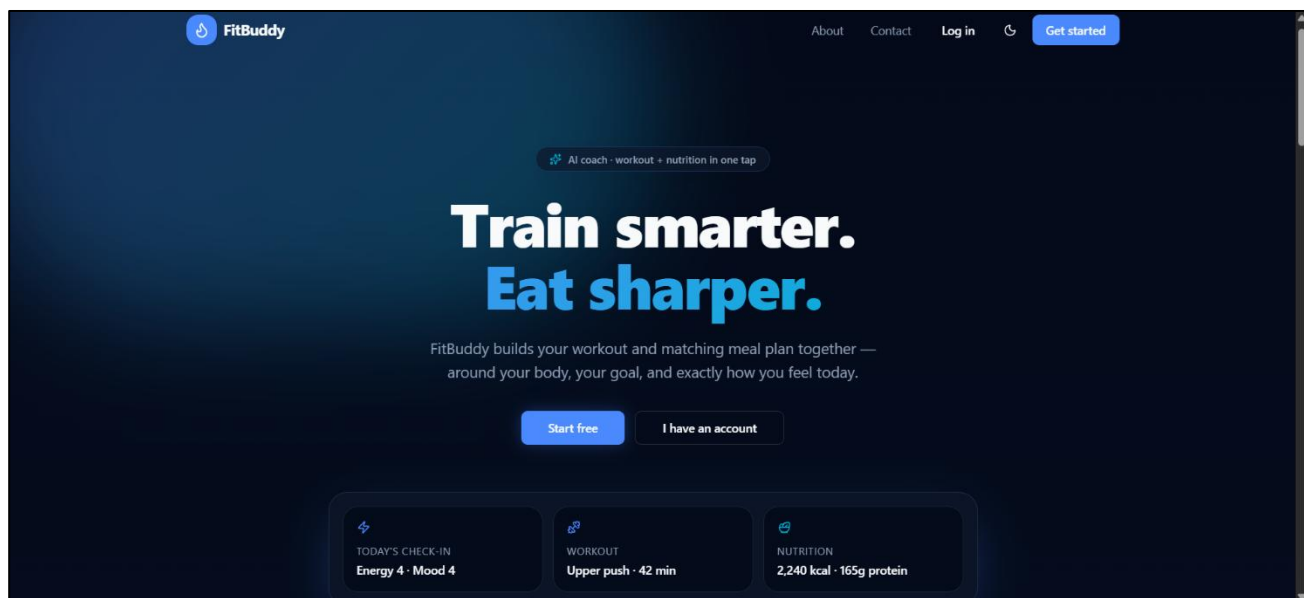
- Enhanced system reliability.
- Increased security.
- Stable AI recommendations.
- Accurate fitness tracking.



## 8. SNAPSHOTS

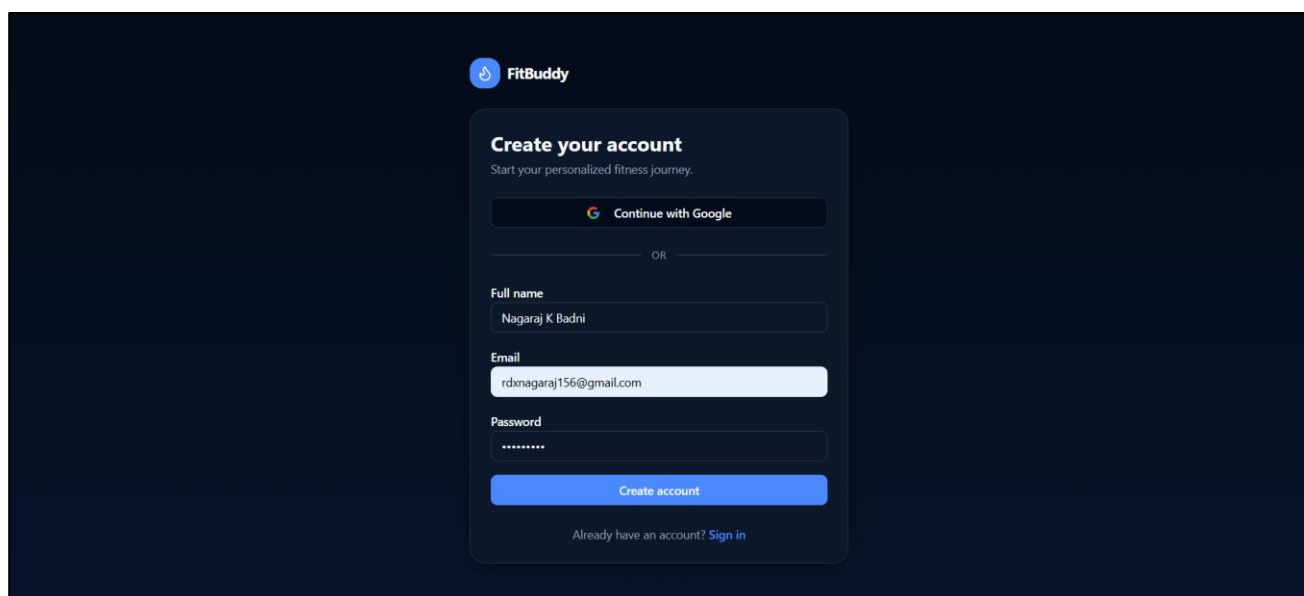
This chapter contains screenshots of the developed application. These screenshots demonstrate the implementation of various modules and features of the FitBuddy system.

### 8.1 Home Page



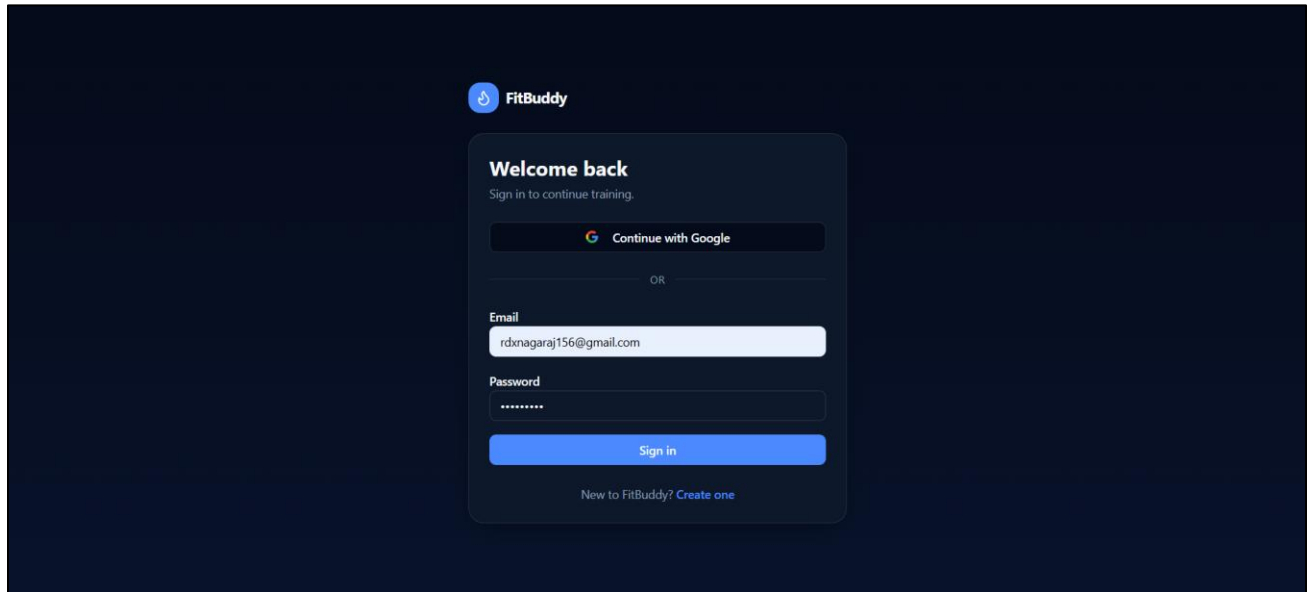
**Description:** The Home Page introduces the FitBuddy application and provides information about its features. Users can navigate to Login or Register to access the platform.

### 8.2 User Registration



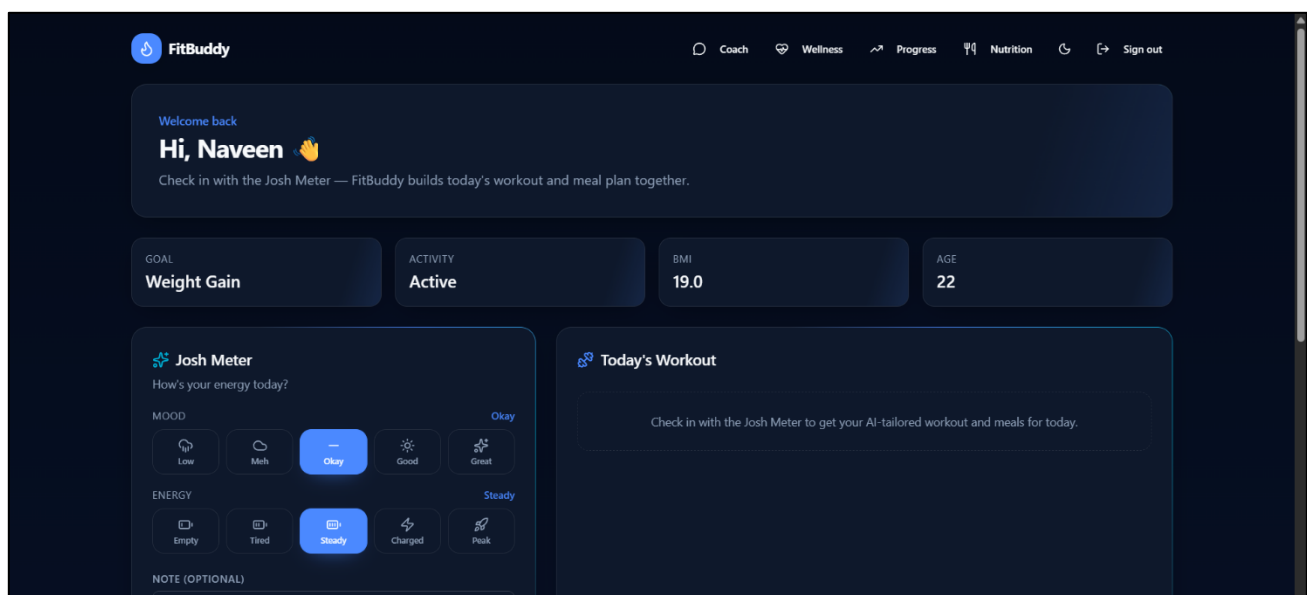
**Description:** The Registration page allows new users to create an account by entering their personal details.

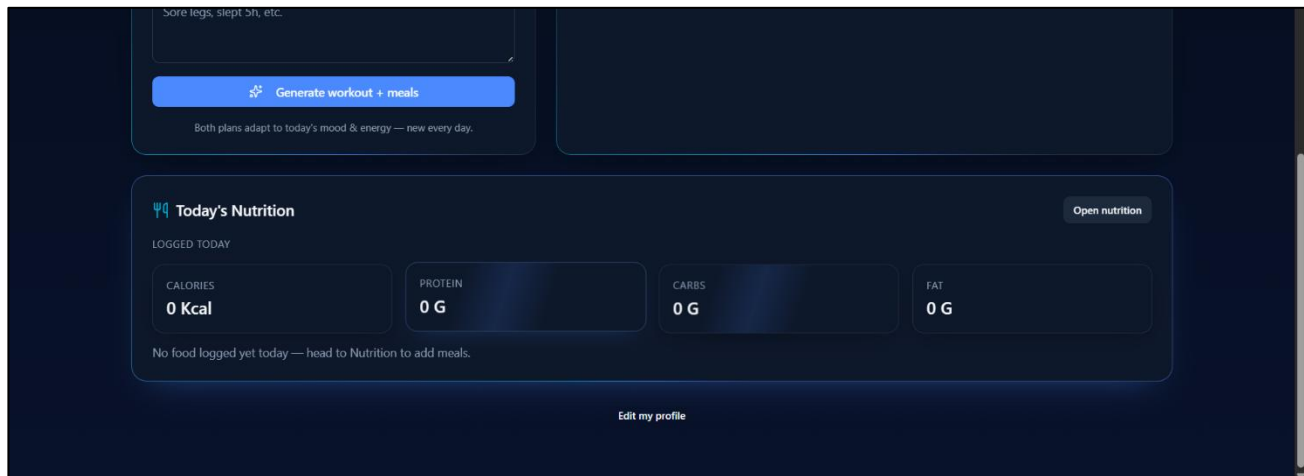
### 8.3 Login Page



**Description:** The Login page allows registered users to securely access the application using Email/Password or Google Authentication.

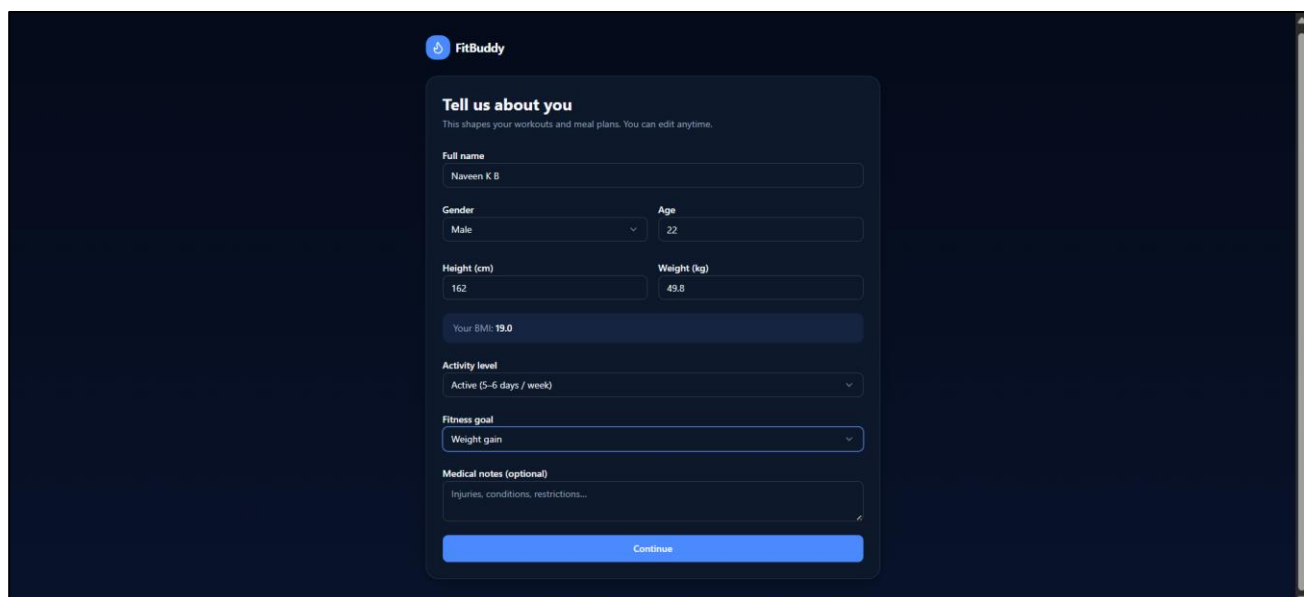
### 8.4 Dashboard





**Description:** The Dashboard displays an overview of workout plans, nutrition plans, wellness summary, Josh Meter status, and progress reports.

### 8.5 User Profile



**Description:** The Profile page stores personal information such as age, height, weight, BMI, activity level, and fitness goals.

## 8.6 AI Workout Planner

**Josh Meter**  
How's your energy today?

MOOD: Low, Med, Okay, **Good**, Great

ENERGY: Empty, Tired, Steady, Charged, **Peak**

NOTE (OPTIONAL): Sore legs, slept 5h, etc.

[Regenerate today's plan](#)

Both plans adapt to today's mood & energy — new every day.

**Today's Workout**  
Full Body Strength  
Intense 60 min 8 exercises

**Barbell Back Squat**  
Focus on depth and controlled movement. 4x8-8 • rest 90-120 seconds

**Incline Dumbbell Press**  
Squeeze at the top, control the eccentric. 3x8-10 • rest 60-90 seconds

**Bent-Over Barbell Row**  
Maintain a flat back, pull with your lats. 3x8-10 • rest 60-90 seconds

**Overhead Press (Dumbbells or Barbell)**  
Keep core tight, press directly overhead. 3x8-10 • rest 60-90 seconds

**Romanian Deadlift**  
Feel the stretch in your hamstrings, don't round your back. 3x8-10 • rest 60-90 seconds

**Plank**  
Keep your body in a straight line, engage your core. 45-60 seconds

**Bicep Curls**  
Controlled movement, focus on the squeeze. 3x10-12 • rest 60 seconds

**Tricep Pushdowns**  
Full extension, keep elbows tucked in. 3x10-12 • rest 60 seconds

**Coach's note**  
Naveen, with your excellent energy today, we're going for an intense full-body strength session to help you achieve your weight gain goals! Focus on progressive overload and proper form. You've got this!

**Description:** This module generates personalized workout schedules based on the user's fitness goals and motivation level.

## 8.7 AI Nutrition Planner

**FitBuddy** ← Dashboard

**Nutrition**  
**Today's plate**  
Generated from today's mood and energy — alongside your workout. Log what you actually eat below.

[Regenerate meal plan](#) [Check in on dashboard](#)

**Your meal plan**  
**Naveen K B's Meal Plan - 2026-08-06**

| KCAL | P    | C    | F   |
|------|------|------|-----|
| 2920 | 146g | 365g | 97g |

**Breakfast** **Hearty Breakfast Burrito** 600 kcal • P 30g • C 60g • F 30g  
3 whole eggs, 1/2 cup black beans, 1/4 cup shredded cheese, 1/2 avocado, 1 whole wheat tortilla, salsa to taste

**Lunch** **Chicken and Quinoa Power Bowl** 750 kcal • P 45g • C 75g • F 35g  
4 oz grilled chicken breast, 1 cup cooked quinoa, 1 cup mixed greens, 1/2 cup roasted sweet potato, 1/4 cup chopped walnuts, 2 tbsp olive oil vinaigrette

**Food log**

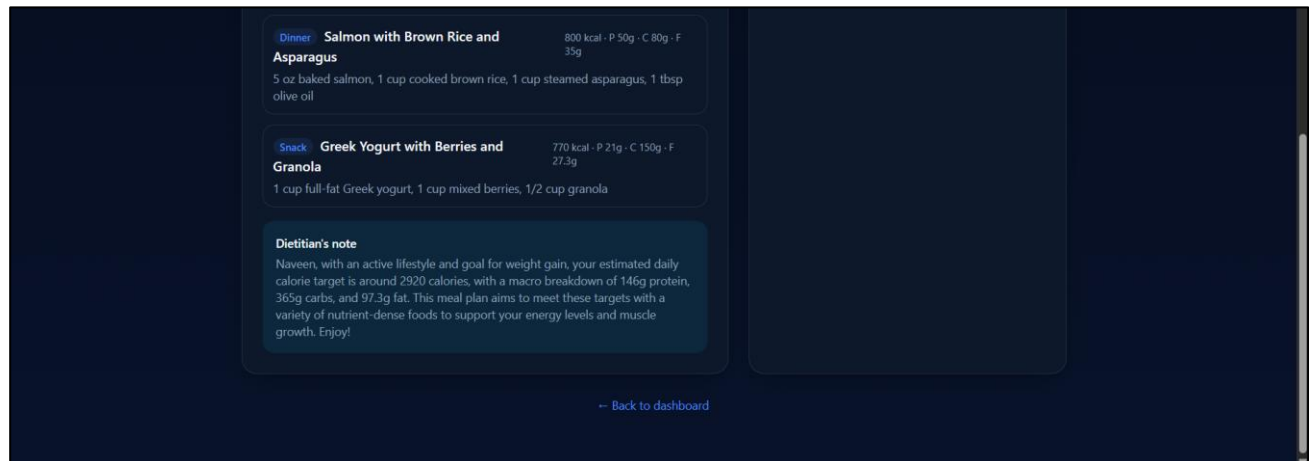
| KCAL | P  | C  | F  |
|------|----|----|----|
| 0    | 0g | 0g | 0g |

Meal: Breakfast Food: Oats & banana

| kcal | P | C | F |
|------|---|---|---|
|      |   |   |   |

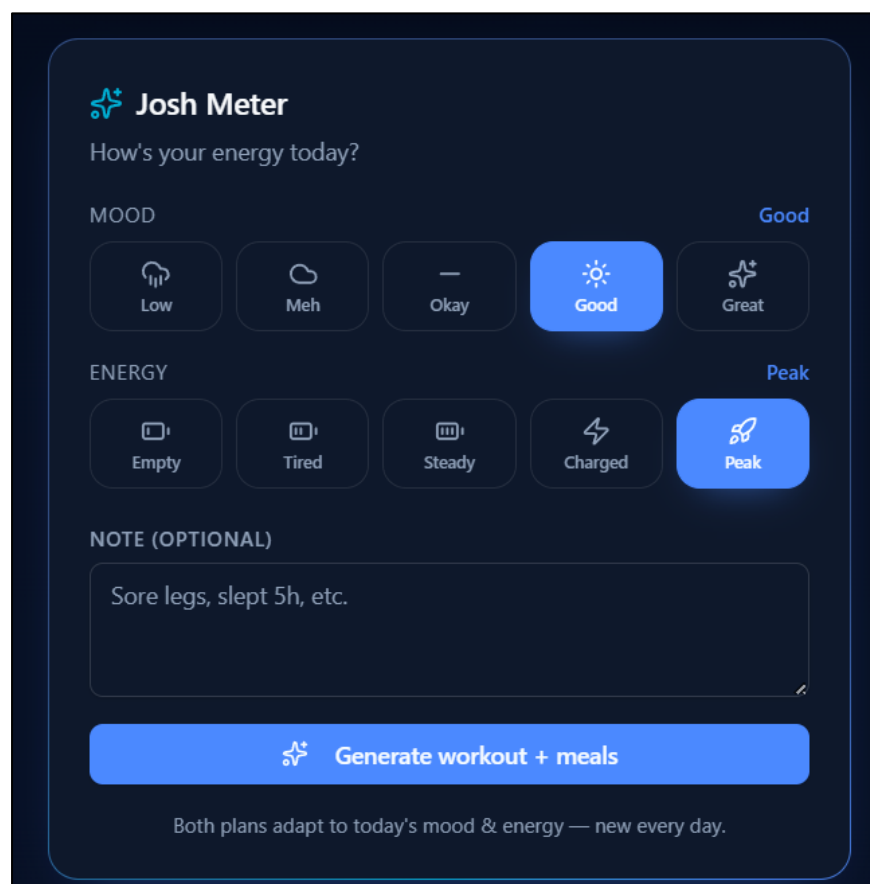
[+ Log food](#)

Nothing logged yet today.



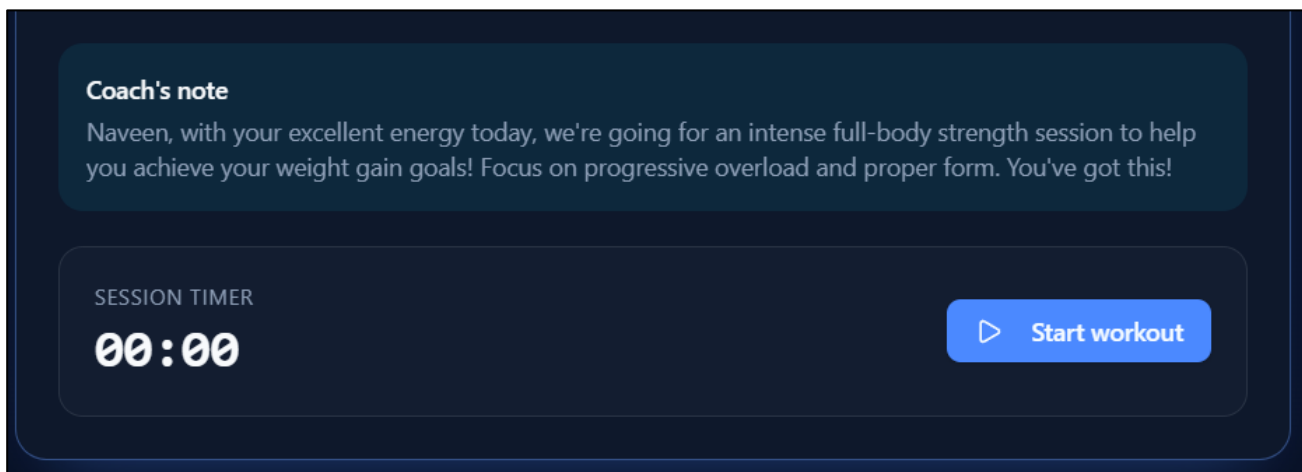
**Description:** The Nutrition Planner provides AI-generated meal recommendations and calorie intake suggestions.

### 8.8 Josh Meter



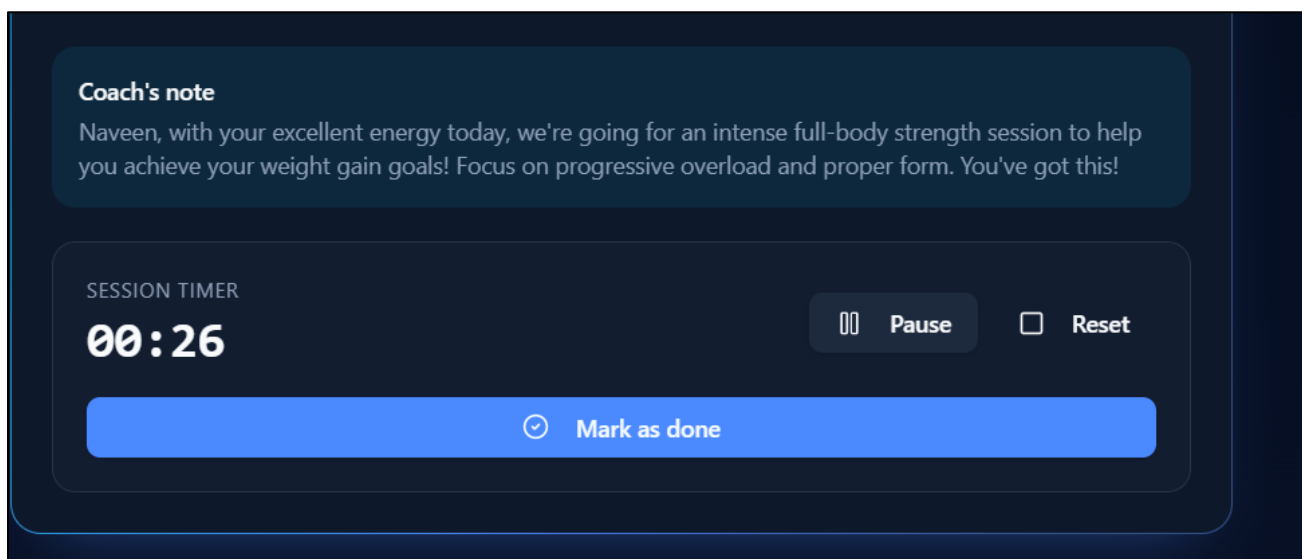
**Description:** The Josh Meter measures the user's daily motivation level and adjusts workout intensity accordingly.

## 8.9 Start Workout



**Description:** Users can start guided workout sessions and monitor exercise progress.

## 8.10 Workout Timer



**Description:** The Workout Timer helps users manage exercise duration and rest intervals effectively.

## 8.11 Wellness Tracker

The screenshot displays the 'Wellness' section of the FitBuddy app. It features two main tracking cards: 'Hydration' and 'Sleep'. The 'Hydration' card shows a progress bar for daily water intake, currently at 4000 ml out of a 2500 ml goal (100%). Below the bar are buttons to add 250 ml, 500 ml, or 750 ml, a 'Reset' button, and a 'Custom ml' input field with an 'Add' button. The 'Sleep' card prompts the user to 'Log last night's rest'. It includes a 'Hours slept' input field set to 8, a 'Quality' selector with buttons 1 through 5 (button 4 is selected), and a 'Notes (optional)' text area containing the text 'Woke up refreshed / restless / etc.'. A 'Save sleep' button is located below the notes. At the bottom, a 'Last 1 nights' summary shows the date '2026-08-06', '8.0h' of sleep, and 'quality 4/5'.

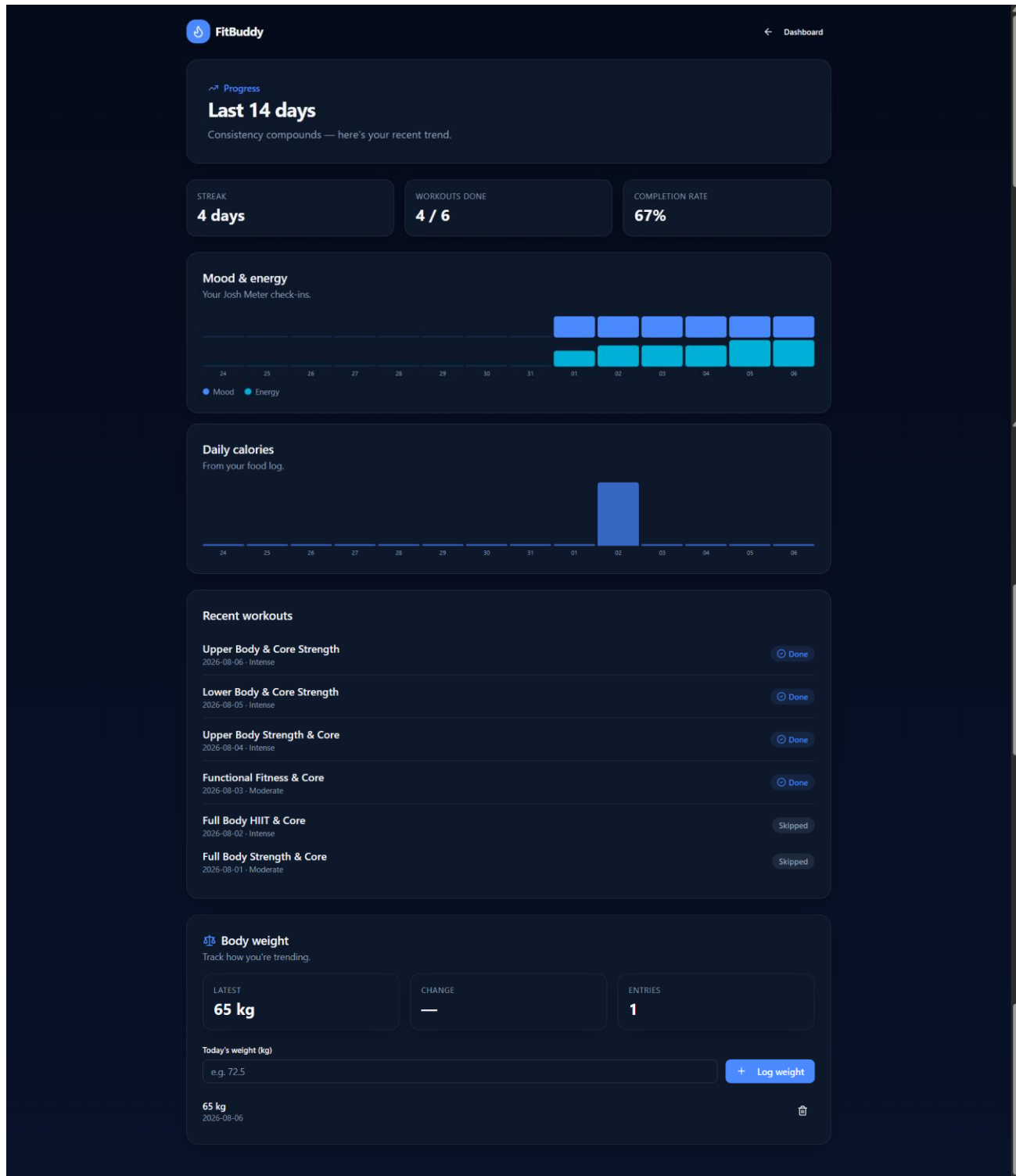
**Description:** The Wellness Tracker records sleep, water intake, mood, weight, and BMI to promote healthy habits.

## 8.12 AI Coach

The screenshot shows the 'AI Coach' interface, titled 'AI Fitness Coach'. It features a chat window with a message from the AI coach: 'Hey! I'm FitBuddy, your AI coach. Ask me anything about training, nutrition, recovery, or motivation.' Below this is a blue button labeled 'Suggest a 20-minute hotel-room workout'. The chat continues with a response from the user: 'Hey Naveen! Great to hear from you. Let's get you a solid hotel room workout.' The AI coach then provides a detailed 20-minute, no-equipment hotel room workout plan. The plan includes a warm-up (3 minutes) with arm circles, leg swings, torso twists, and light marching. It also includes a workout circuit (15 minutes - 3 rounds, 1-minute rest between rounds) with bodyweight squats. At the bottom, there is a text input field labeled 'Ask your coach anything...' and a blue button with a paper plane icon.

**Description:** The AI Coach provides personalized fitness guidance, workout advice, and nutrition suggestions using Gemini AI.

## 8.13 Progress Tracker





**Description:** The Progress Tracker displays workout history, wellness statistics, and fitness improvements using charts and reports.

## **9. CONCLUSION**

FitBuddy – AI-Powered Personal Workout Coach & Adaptive Nutrition Planner is developed to provide users with an intelligent and personalized fitness platform. The application combines Artificial Intelligence, personalized workout planning, adaptive nutrition recommendations, wellness tracking, and AI coaching into a single system.

The application successfully generates workout schedules and nutrition plans based on the user's profile, fitness goals, and daily motivation level through the Josh Meter. Additional features such as Start Workout, Workout Timer, Wellness Tracker, and AI Coach improve user engagement and help users maintain consistency in their fitness journey.

The responsive interface, secure authentication, and modern technology stack make the application reliable and easy to use. Overall, FitBuddy provides an effective solution for individuals seeking personalized fitness guidance without requiring a personal trainer.

### **Project Outcomes**

- Personalized AI-based workout recommendations.
- Adaptive nutrition planning.
- Daily motivation analysis using Josh Meter.
- Guided workout sessions with timers.
- Wellness monitoring.
- AI-powered fitness coaching.
- Progress tracking and reporting.
- Improved user engagement and healthy lifestyle management.

## **10. FUTURE ENHANCEMENTS**

The FitBuddy application can be enhanced further by incorporating additional technologies and features to improve user experience and functionality.

Future enhancements include:

- Integration with wearable devices such as smartwatches and fitness bands.
- Real-time heart rate and step-count monitoring.
- Voice-based AI fitness coach.
- Exercise posture detection using computer vision.
- Barcode scanning for food nutrition analysis.
- AI-based injury prevention recommendations.
- Social fitness community and challenge system.
- Video-based exercise demonstrations.
- Multi-language support.
- Mobile application for Android and iOS.
- Offline workout mode.
- Personalized health analytics dashboard.

These enhancements will make FitBuddy more comprehensive, intelligent, and suitable for a wider range of users.

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